

Generalized linear models

Logistic regression

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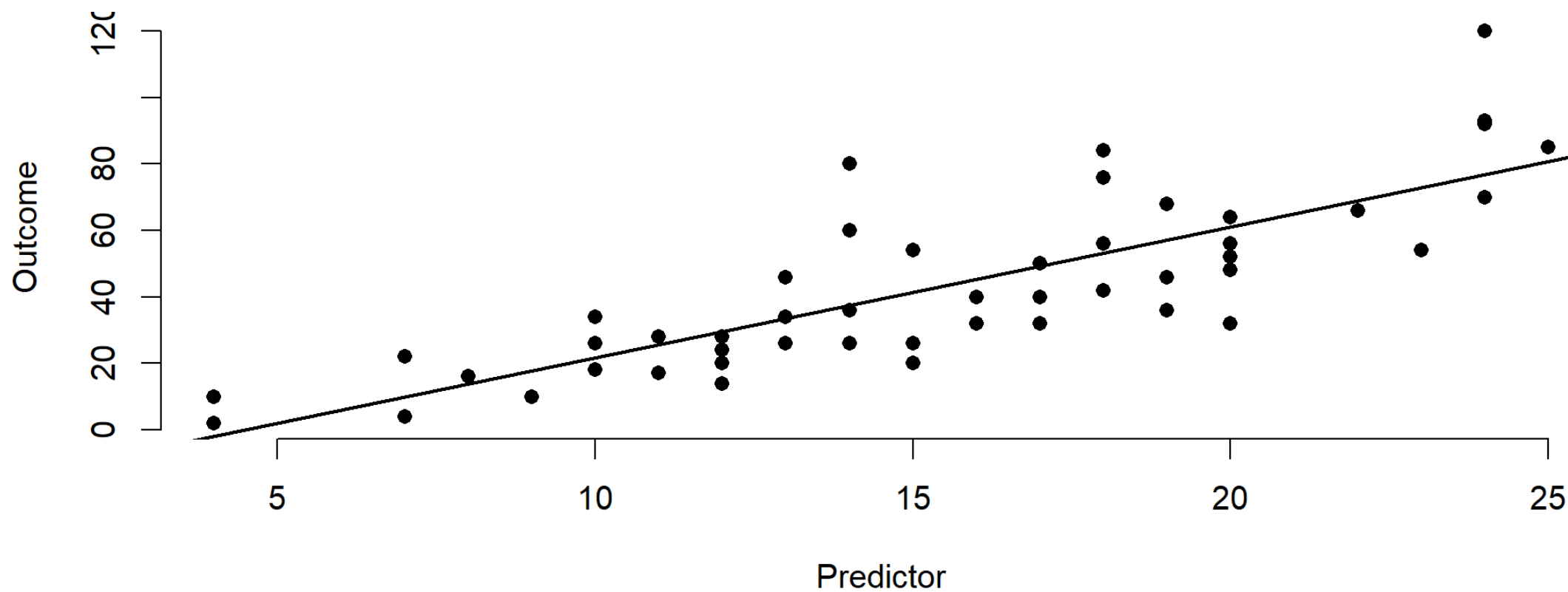
Press record

First meeting practice

Linear regression

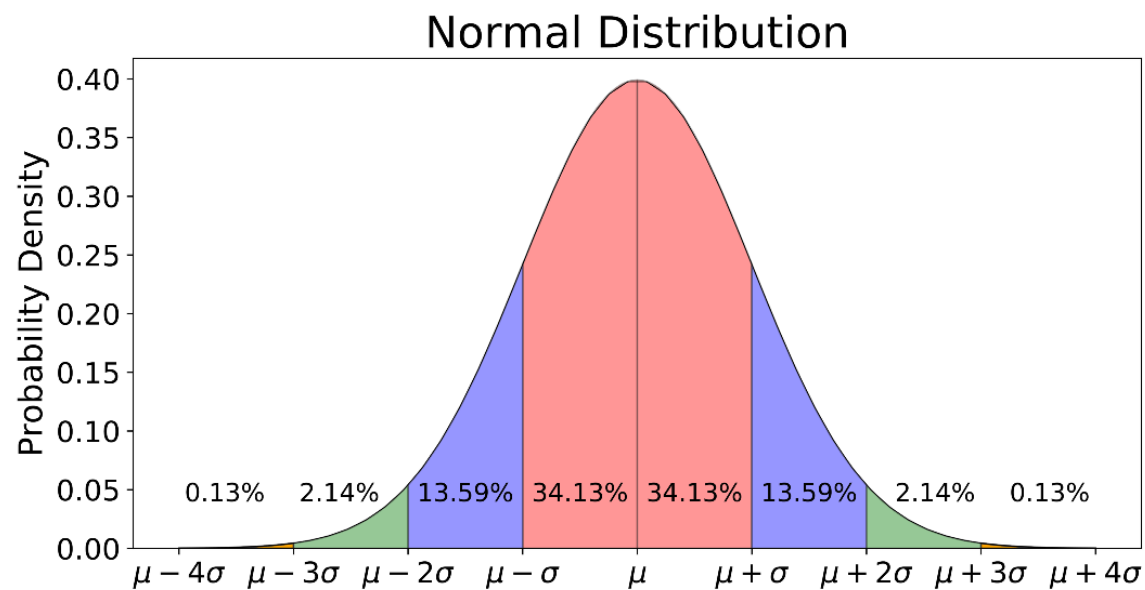
Gaussian distribution: $y_i \sim \mathcal{N}(\mu_i, \sigma)$

$$\mu_i = \alpha + \beta * x_i$$



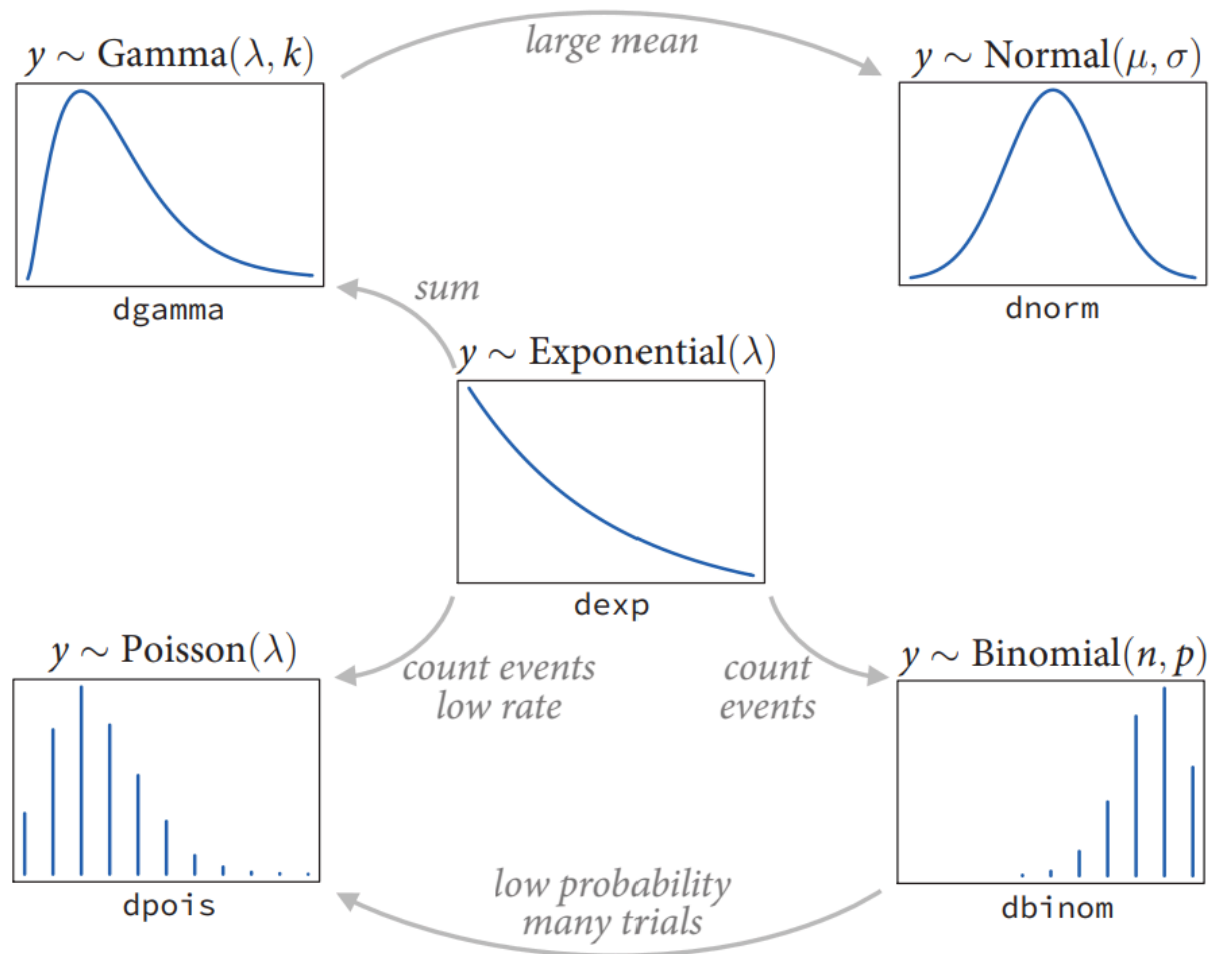
Gaussian distribution

$$y \sim \mathcal{N}(\mu, \sigma)$$



Can you tell me about other probability distributions?

Exponential family



Binomial distribution

Distribution:

$$y_i \sim \text{Binomial}(n, p)$$

Number of successes of n number of independent Bernoulli trials -> Binomial distribution:

Probability mass function: $P(X = x) = \binom{n}{x} p^x * (1 - p)^{n-x}$

Mean: $\mu = n * p$

Variance: $\sigma^2 = n * p(1 - p)$

Logistic regression

$$p = \alpha + \beta * x_i$$

What would happen if we would model probability using linear regression?

Linear function is unbounded - does not have to result in a number between 0 and 1

Errors are not necessarily normal

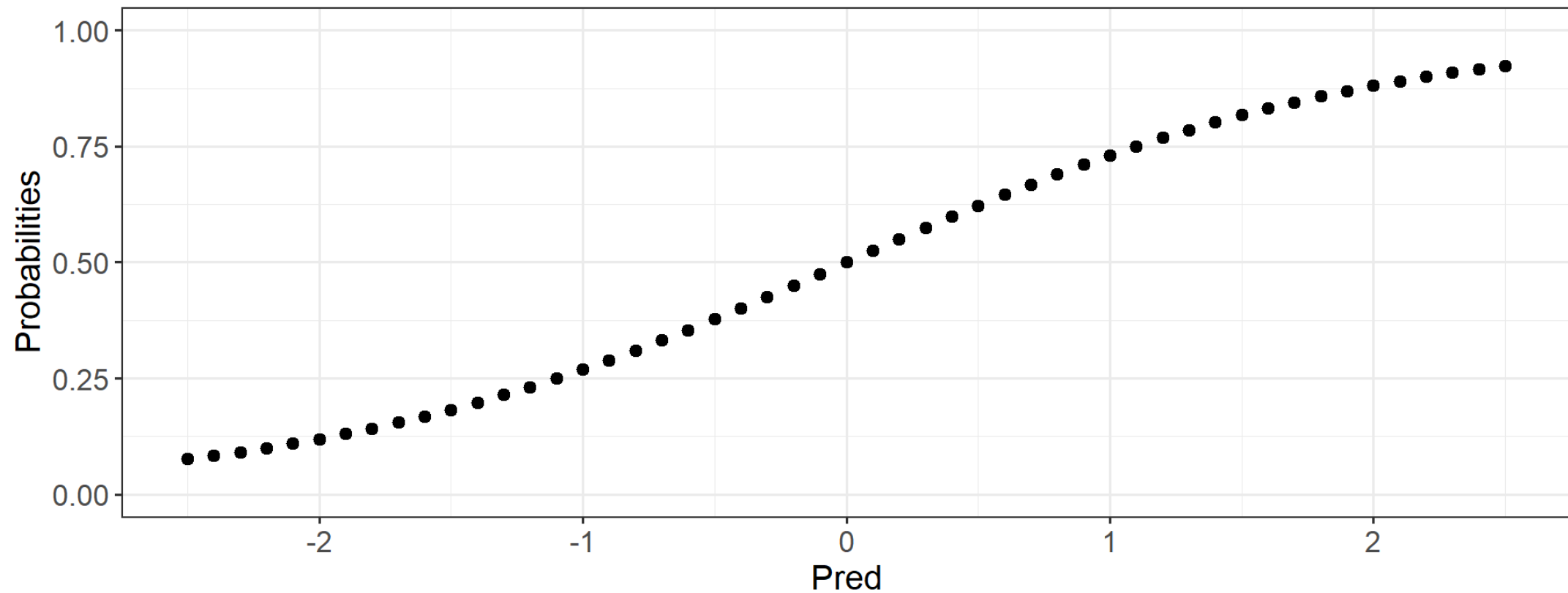
Mean and SD are related

Map values to [0,1]:

$$f(p) = \log\left(\frac{p}{1-p}\right)$$

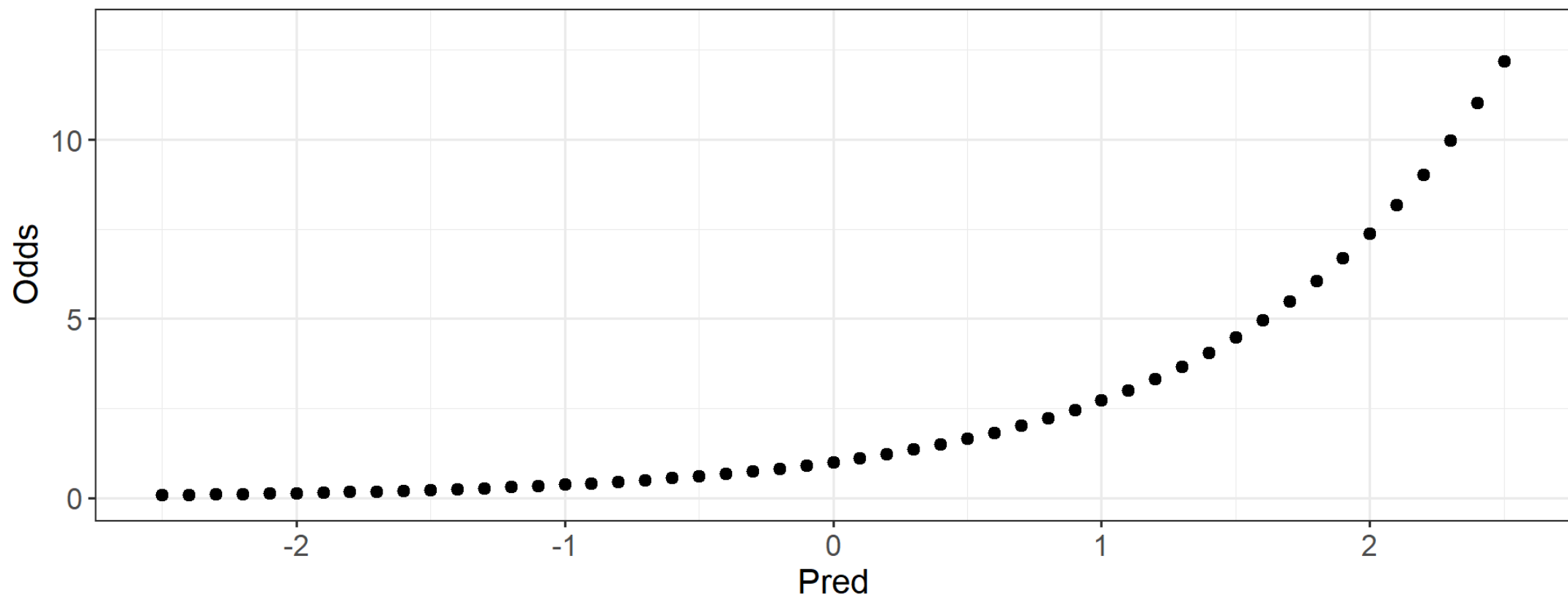
$$\text{logit}(p_i) = \alpha + \beta * x_i$$

Probabilities



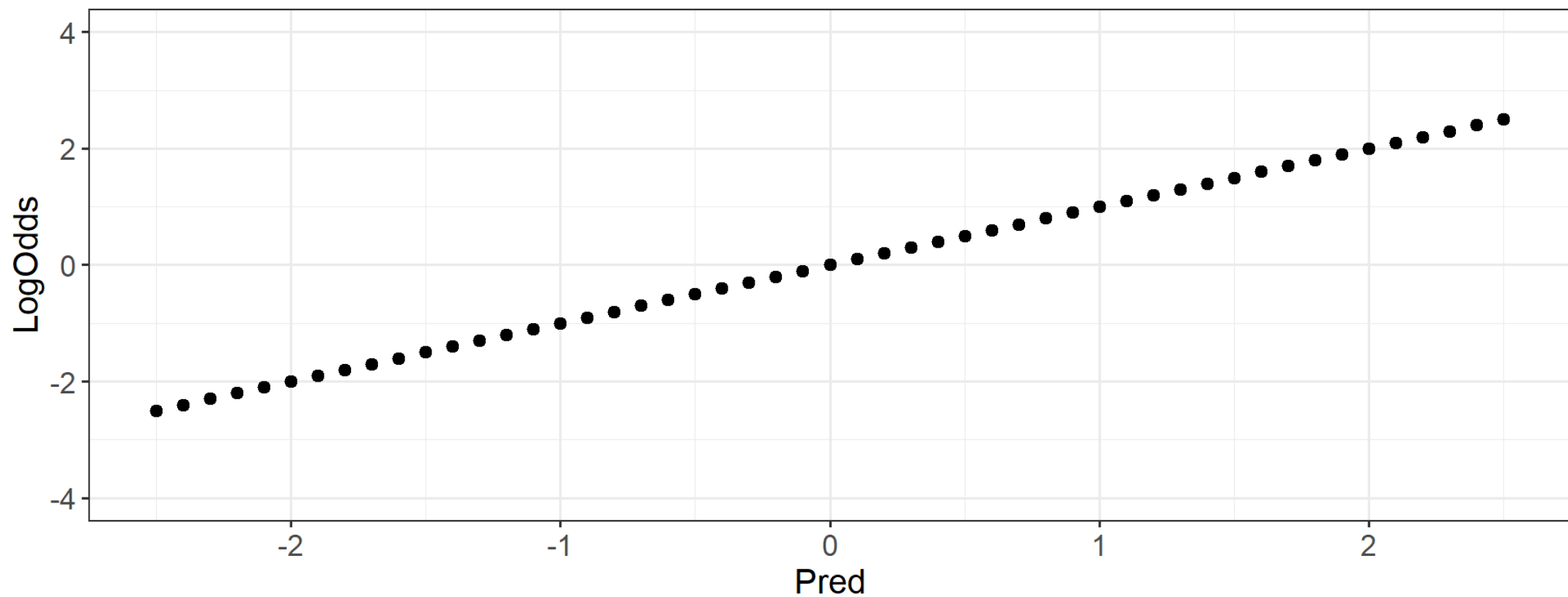
First step: odds

$$Odds = \frac{p_i}{1 - p_i}$$

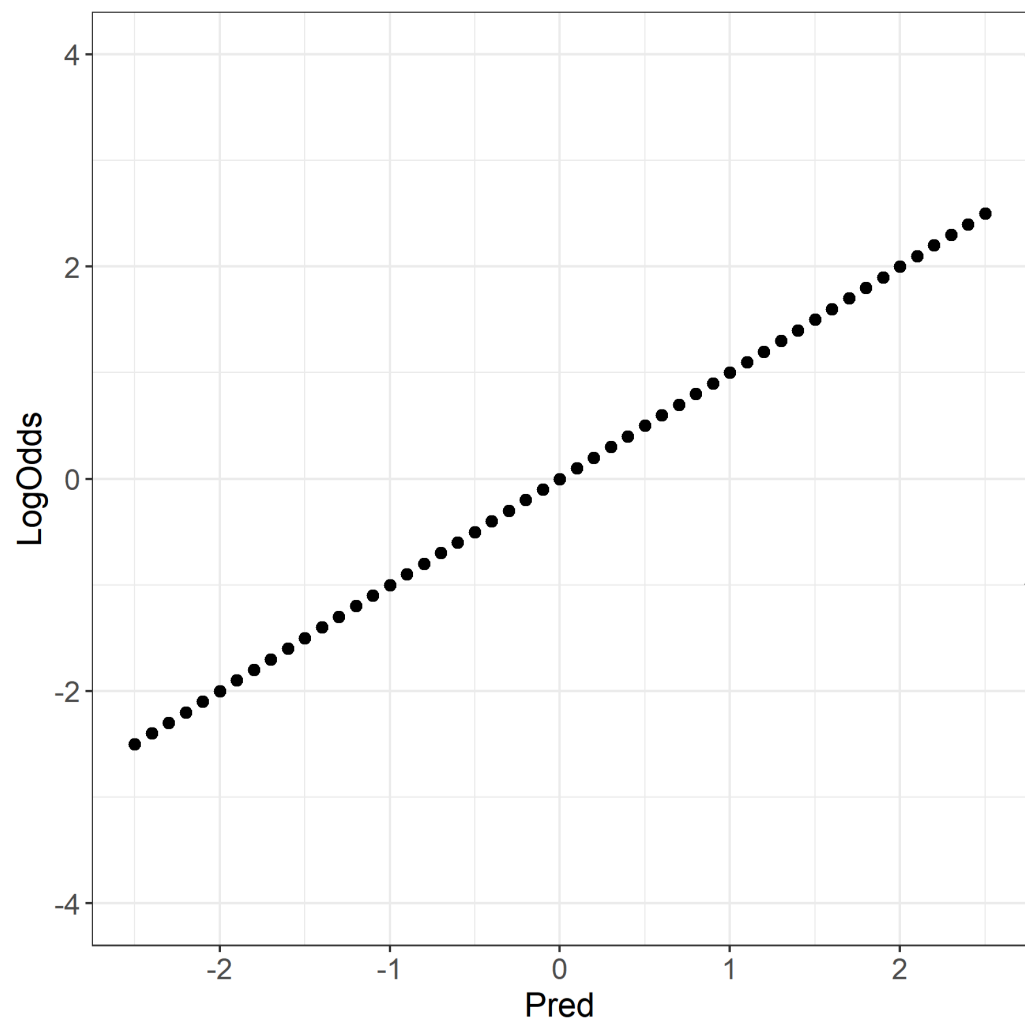


Second step: log odds

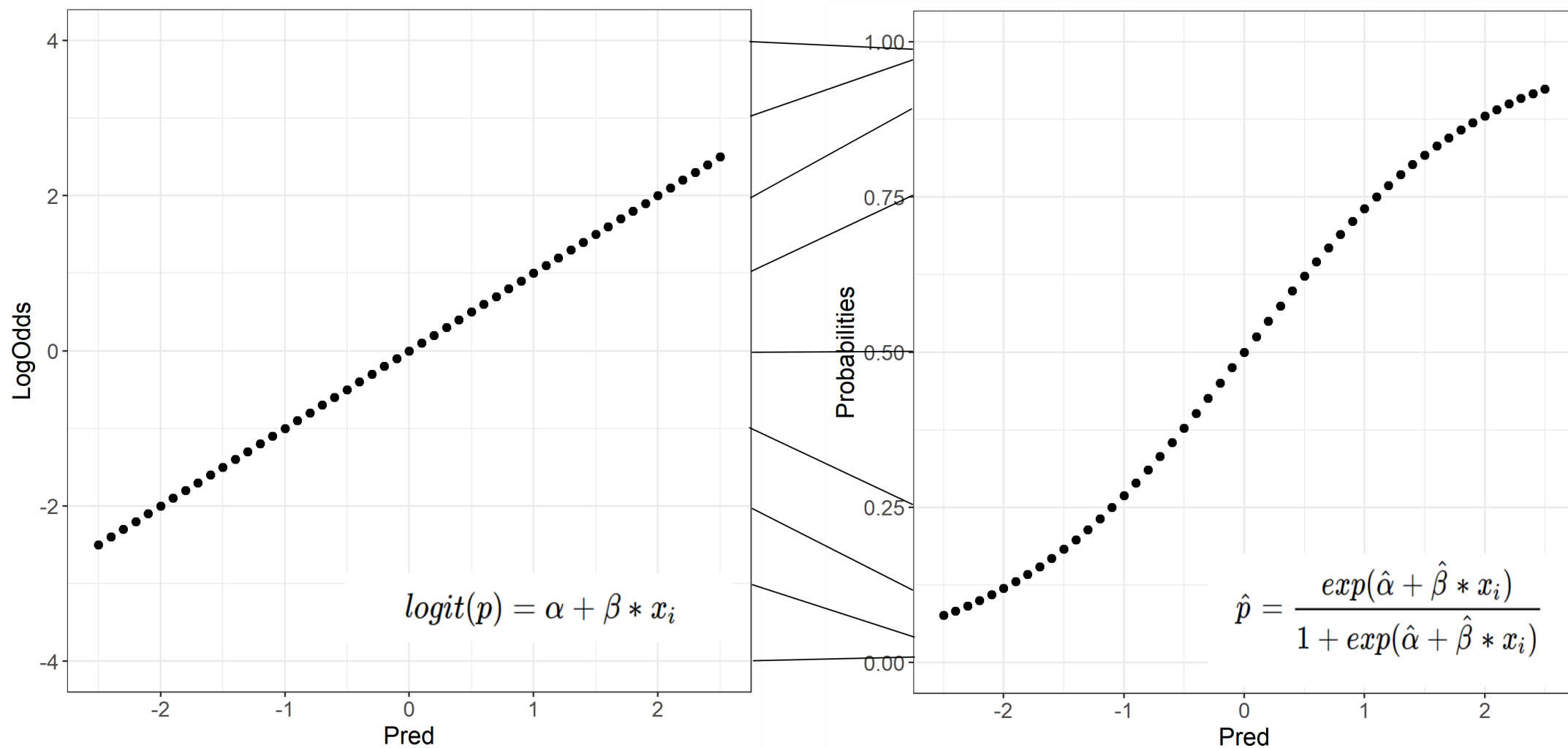
$$\log Odds = \log\left(\frac{p_i}{1 - p_i}\right)$$



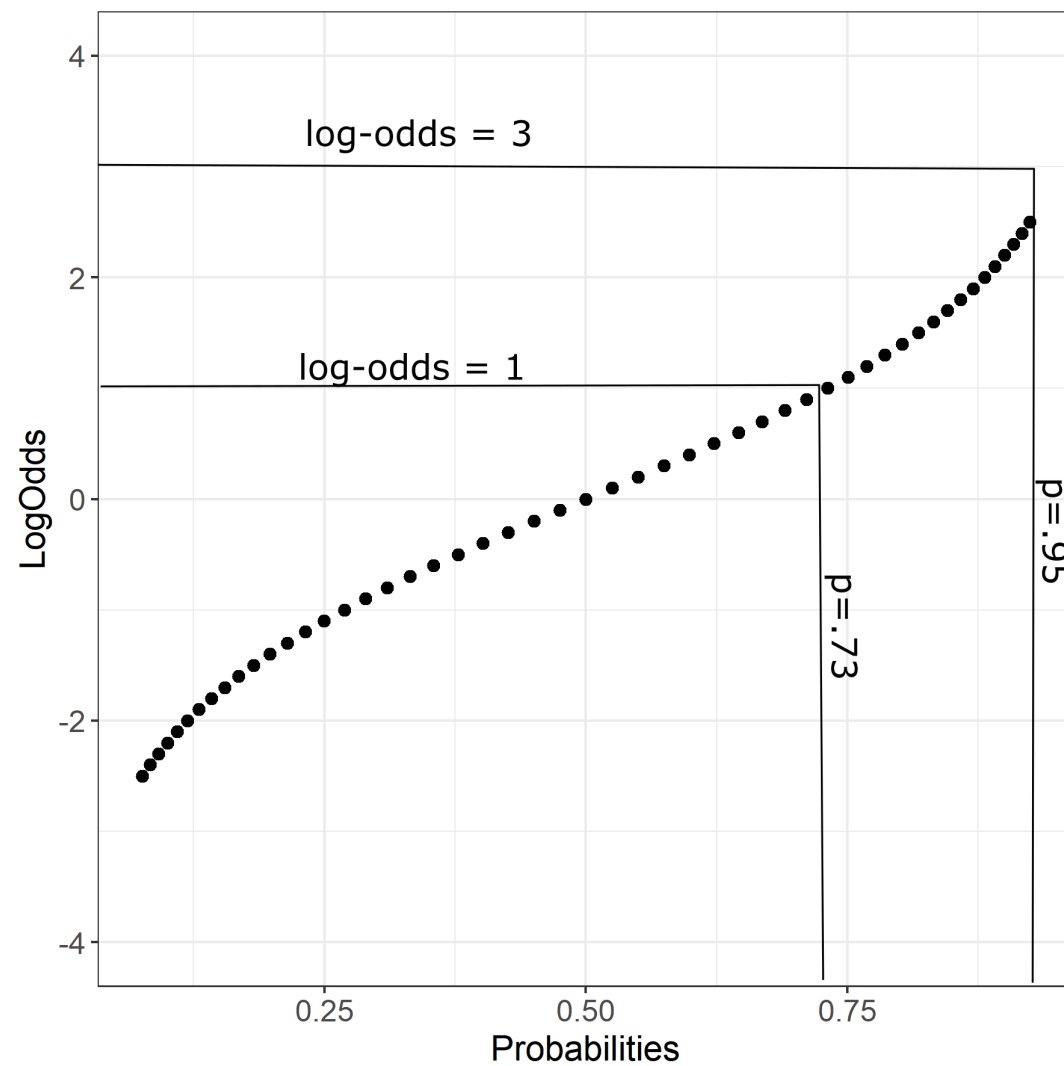
What is the model doing: log odds space?



What is the model doing: probability space?

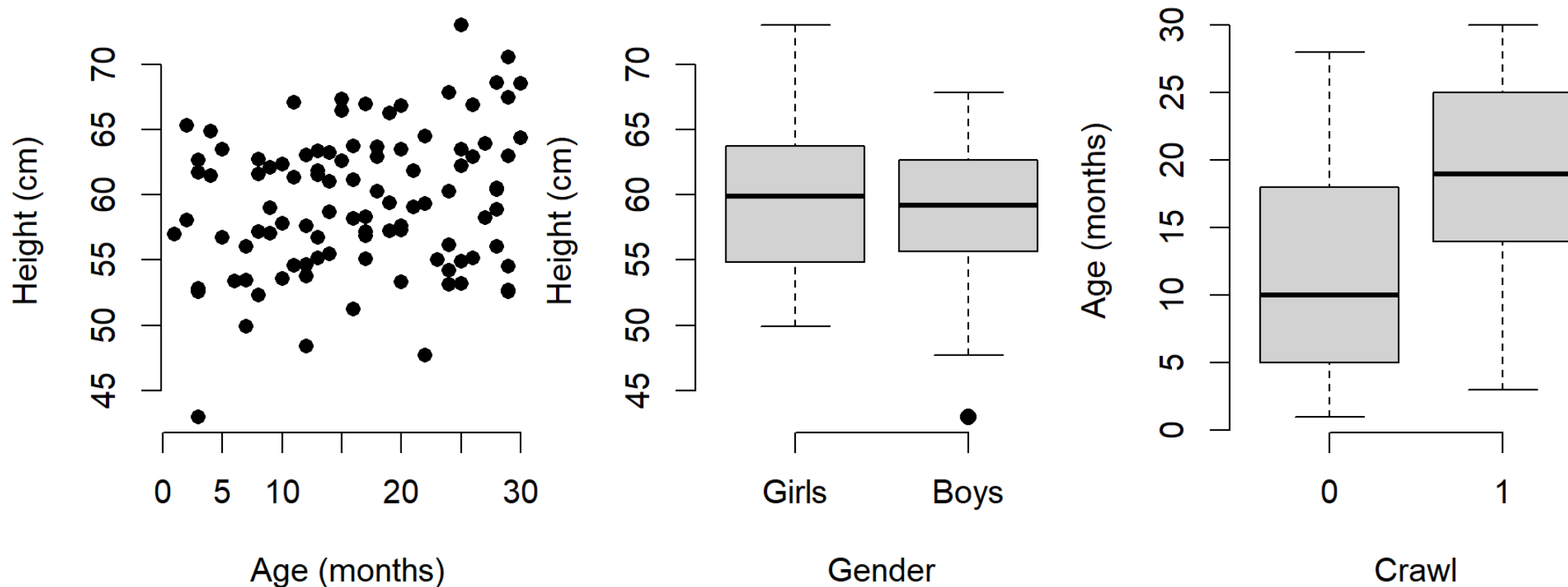


Relation between LogOdds and Probabilities



Logistic regression in R

We return back to our crying babies that are motivated to start crawling.



Logistic regression in R: coefficients

Predict crawling success as a linear function of age

```
glm1 ← glm ( Crawl ~ Age, data = Babies, family = binomial ( link = 'logit' ))  
glm1 $ coefficients
```

```
## (Intercept)          Age  
##   -1.3307752    0.1194777
```

Three ways to interpret logistic regression:

1. Logit values
2. **Odds**
3. **Probabilities**

Odds ratio space

Odds:

```
exp ( glm1 $ coefficients )
```

```
## (Intercept)          Age  
##    0.2642723    1.1269081
```

Intercept: When Age is **zero**, the **odds** of babies crawling are 74% (1-.26) **less likely** than odds of crawling.

Age: Babies **one month older** are 12% more likely to start crawling OR unit increase in Age **increases the odds** of crawling by a factor of 1.12

Probability space: intercept

Logit to probability space:

$$\frac{1}{1 + \exp^{-(\alpha + \beta * x)}}$$

Intercept: Estimated probability of babies starting to crawl when Age is 0 (birth)

```
1 / ( 1 + exp ( - ( -1.33078 + 0.119 * 0)))
```

```
## [1] 0.2090304
```

Probability space: slope

We need to look where to evaluate changes on the curve:

```
1 / ( 1 + exp( 1.33078 - 0.11948 * 10 ))
```

```
## [1] 0.4660573
```

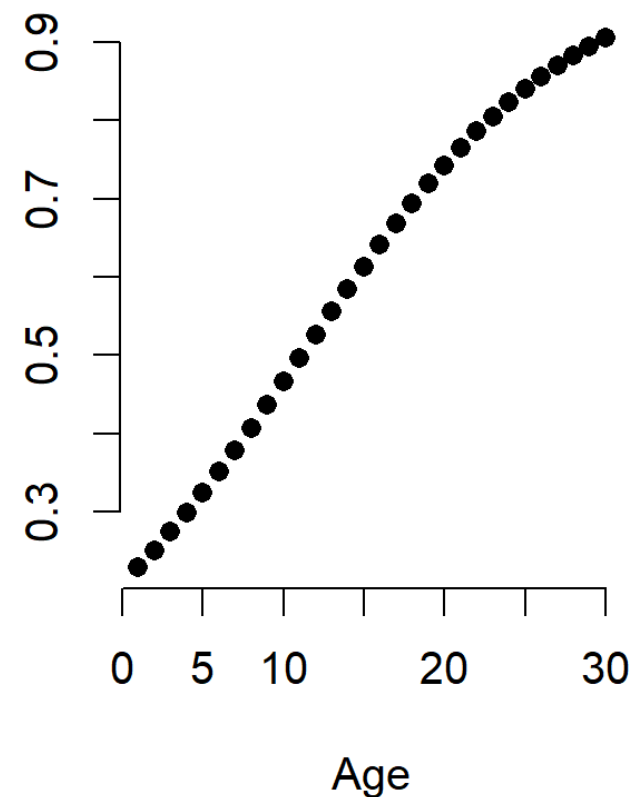
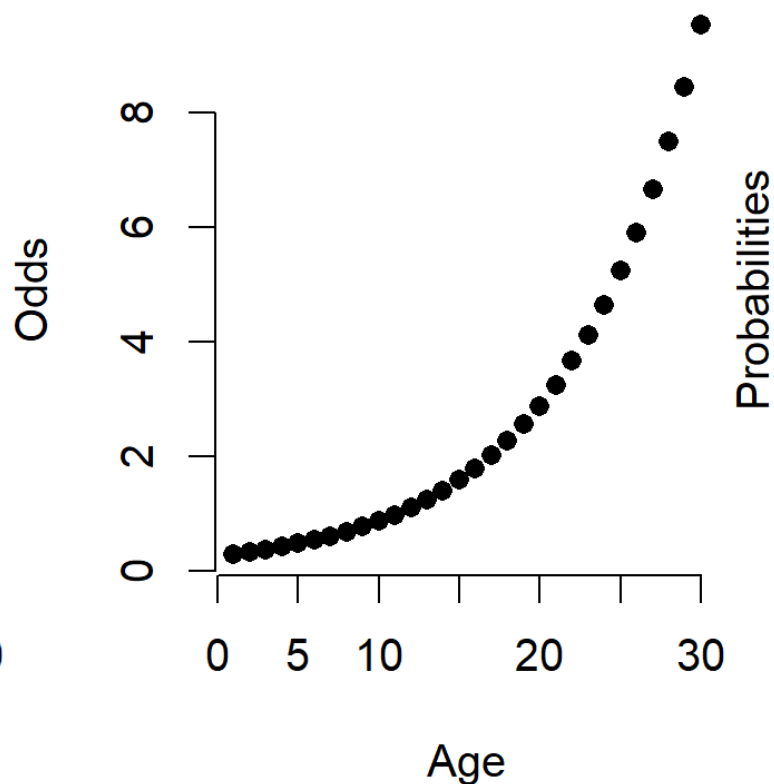
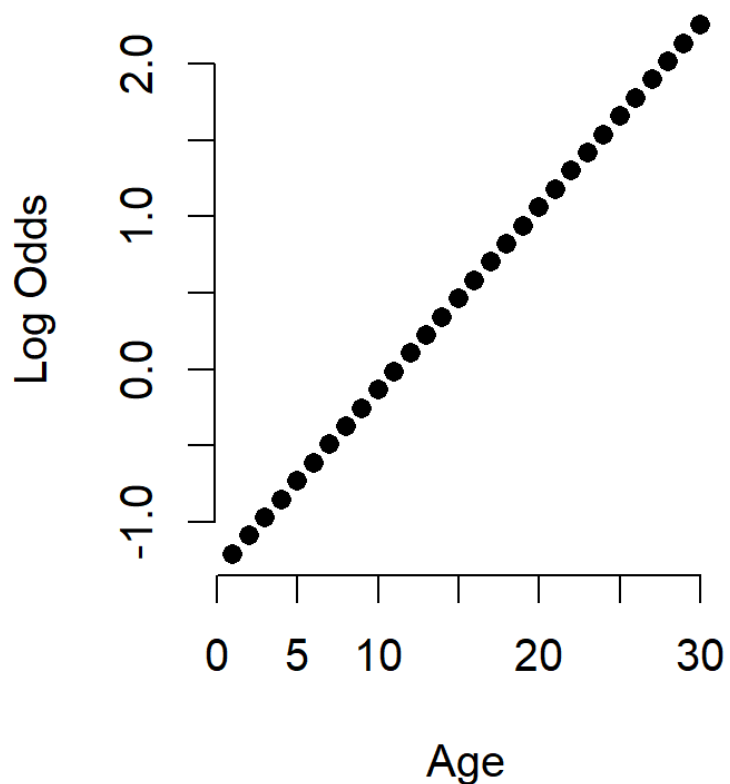
```
arm::invlogit(coef( glm1 )[[ 1 ]] + coef( glm1 )[[ 2 ]] * mean( Babies $ Age ))
```

```
## [1] 0.6562395
```

Estimated probability of babies starting to crawl when Age is 10 or mean of the sample

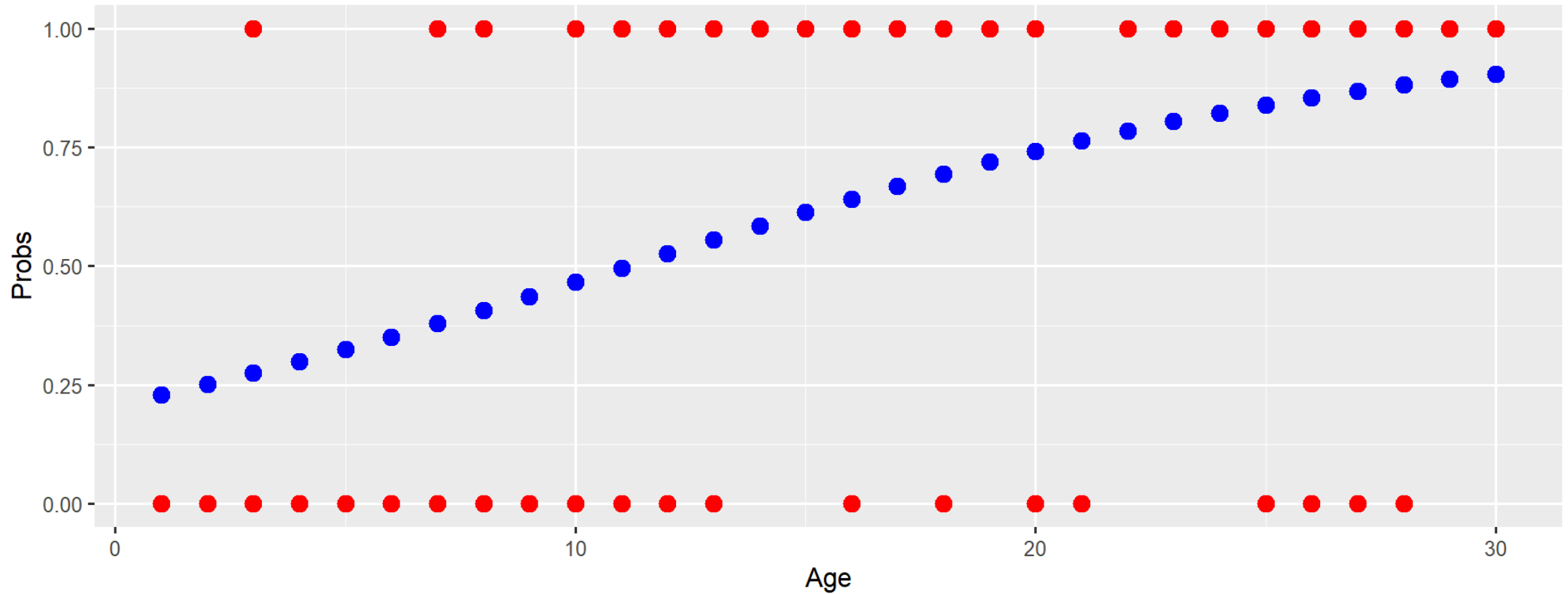
Interpretation continued

```
Babies$LogOdds=-1.33078+0.11948*Babies$Age  
Babies$Odds=exp(Babies$LogOdds)  
Babies$Probs=Babies$Odds/(1+Babies$Odds)
```



Predicted probability versus outcome

```
ggplot(data=Babies, aes(x=Age, y=Probs))+geom_point(size=3, col='blue')+geom_poin
```



Relationship between probabilities and odds

Probability: $p = \frac{f}{N}$

Expectations of how many times we will see event in N number of trials (between 0 and 1)

Odds of event: $odds = \frac{p}{1-p}$ (between 0 and positive infinity)

Probability that the event will occur divided by the probability that event will not occur

$p = 0.80$

$odds = \frac{0.80}{0.20} = 4$ - odds are 4 to 1 that the event will occur

$p = 0.20$

$odds = \frac{0.20}{0.80} = 0.25$ - odds are 0.25 to 1 that the event will occur

If a race horse runs 100 races and wins 25 times and loses 75 of times, the probability of winning is $25/100 = 0.25$ or 25%, but the odds of the horse winning are $25/75 = 0.333$ or 1 win to 3 loses

Logistic regression formula

What is missing in this equation?

$$\log\left(\frac{p}{1-p}\right) = \alpha + \beta * x_i$$

$$y = \alpha + \beta * Age + \epsilon$$

We do not have an estimation of error: $y_i \sim \text{Binomial}(n, p)$

We do have residuals - estimate of the fit

$$residual_i = -2 * (y_i - \text{logit}^{-1}(\beta * x_i))$$

Deviance

Goodness of fit: deviance of the fitted model with respect to a perfect model

Null deviance: $D = -2 * \text{loglik}(\beta_0)$

Residual deviance: $D = -2 * \text{loglik}(\beta * x)$

```
with(glm1, pchisq(null.deviance - deviance, df.null - df.residual, lower.tail = F,
```

```
## [1] 1.751679e-05
```

AIC: $-2\text{loglik}(\beta * x) + 2 * npar$

BIC: $-2\text{loglik}(\beta * x) + \log(\text{number of observations}) * npar$

Model results

```
summary(glm1)
```

```
##  
## Call:  
## glm(formula = Crawl ~ Age, family = binomial(link = "logit"),  
##      data = Babies)  
##  
## Deviance Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.0687  -0.9876   0.5443   0.8979   1.6082   
##  
## Coefficients:  
##
```

Link functions and other GLMs

Most common Generalized linear models:

- Poisson regression: traffic, goals in a soccer game
- Negative binomial: same problems but deals with overdispersion
- Gamma regression: reaction times? always positive right skewed data

Absolutely cool things to note

The question that we are asking is can we stratify the probability of a success in outcome using our predictor variables

We are modelling the mean and we do not have an information on individual data points - no error

The scale of parameters is different!

Parameters are multiplicative - coefficient interact, especially in the tails of probability distribution

Not easy to compare the models - especially if they differ in a more than one predictor - deviance criteria gives you information how well you model fits the mean

Practical aspect

Data

NBA dataset: [Link](#)

```
basketball←read.table('Basketball.txt',sep='\t', header=T)
knitr::kable(head(basketball[,c(5,13,18,31,34,43)]), format = 'html')
```

	Home	X3PointShots	FreeThrows.	Opp.3PointShots.	Opp.FreeThrows.	Win
1	Away	13	0.529	0.308	0.818	0
32	Away	6	0.867	0.323	0.750	1
83	Home	8	0.652	0.368	0.769	1
112	Home	12	0.684	0.481	0.941	0
165	Away	7	0.769	0.364	0.652	0
196	Away	7	0.611	0.500	0.650	1

Important aspects: theory

- How do we use linear model with the non-normal outcomes (link functions)
- What are logit values, how do we get to them
- How to interpret logistic regression: logits, odds, and probabilities
- What is deviance of the model and AIC

Important aspects: practice

- Building a logistic model
- Predicting values from logistic model
- Transforming logit values to odds and to probabilities
- Comparing models using deviance, anova function or AIC information
- Cross-tabulating model predictions with observed data

Literature

Chapter 5 of "Data Analysis Using Regression and Multilevel/Hierarchical Models" by Andrew Gelman and Jennifer Hill

Chapter 10 of "Statistical Rethinking: A Bayesian Course with Examples in R and Stan " by Richard McElreath + check out his youtube channel

Jaeger, F. (2008). Categorical data analysis: Away from ANOVAs (transformation or not) and towards logit mixed models. *Journal of Memory and Language*, 59, 434-446.

Song for the end

Link

Thank you for your attention